# Series & Binomial Theorem

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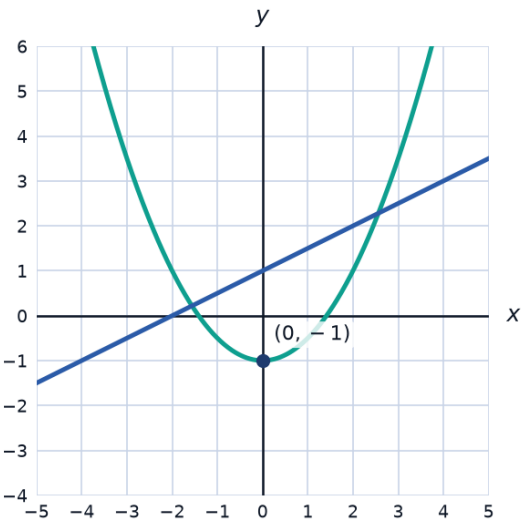
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A particle moves along a straight line. Its velocity, $v$ m/s, at time $t$ seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and $c$ are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and $c$.




**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^nC_r = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?

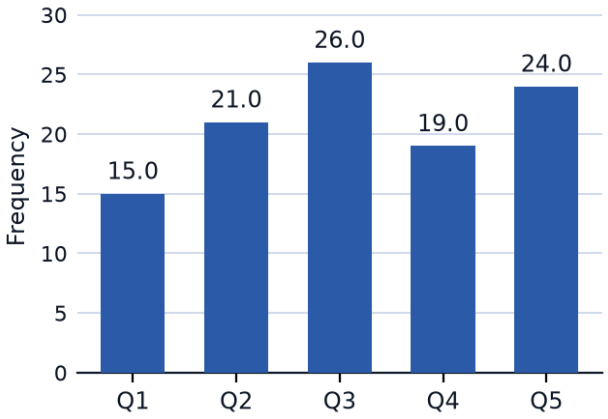
**4** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A particle of mass 3 kg is at rest on a smooth horizontal surface. A horizontal force of magnitude P N is applied to the particle, causing it to accelerate at a rate of 4 m/s$^2$. Find the value of P.

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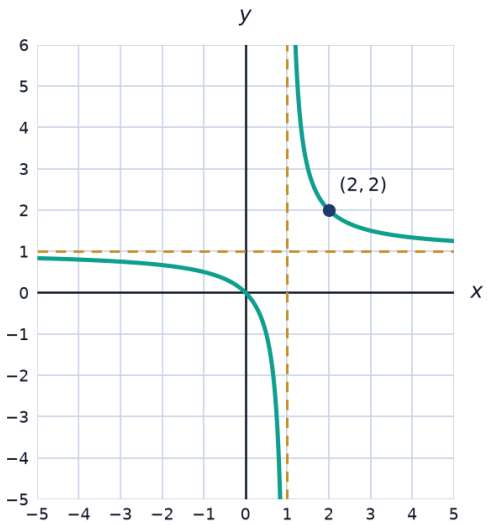
**5** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The two-way table shows the results of a survey of 100 students on their preferred mode of transport to school and their year group. | | Car | Bus | Walk | Total | |----------|-----|-----|------|-------| | Year 1 | 15 | 20 | 10 | 45 | | Year 2 | 25 | 18 | 12 | 55 | | Total | 40 | 38 | 22 | 100 | What is the probability that a randomly selected student is a Year 1 student who prefers to travel by car?




**6** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force $F$ N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.




**7** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A factory tests 30 independent components, each with a fixed probability of being defective; $X$ is the number of defective components. Is $X$ binomially distributed?

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

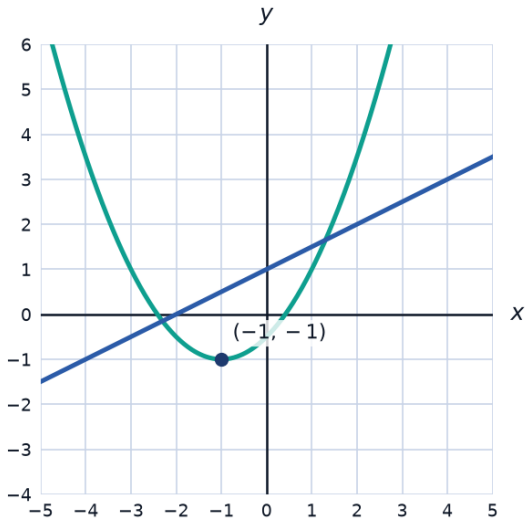
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**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A survey asks 50 randomly and independently selected people the same yes/no question; $X$ is the number who say yes. Is $X$ binomially distributed?

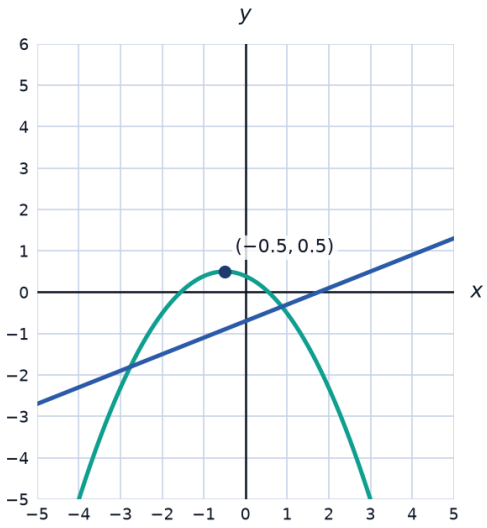
**9** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A machine has a useful power output of $883$ W and wastes $367$ W as heat. Find its efficiency as a percentage, correct to 1 decimal place.




**10** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle of mass $5$ kg slides down a rough plane inclined at $30^\circ$ to the horizontal, experiencing a constant friction force of $0$ N. Taking $g = 9.8$ m/s$^2$, find its acceleration down the slope.




**11** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Given $X \sim N(\mu, \sigma^2)$, $P(X < 99) = 0.8$ and $P(X < 83) = 0.2$, find $\mu$, correct to the nearest integer.

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**12** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle of mass $m$ kg is placed on a rough inclined plane. The plane is inclined at an angle $\alpha$ to the horizontal, where $\tan\alpha = \frac{3}{4}$. The particle is on the point of sliding down the plane. The coefficient of static friction between the particle and the plane is $\mu_s$. Given that the acceleration due to gravity is $g = 9.8$ m/s$^2$, find the value of $\mu_s$.

**13** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In an arithmetic progression, the 3rd term is $14$ and the 7th term is $34$. (a) Find the first term $a$ and the common difference $d$. (b) Find the sum of the first $25$ terms.

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Series & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

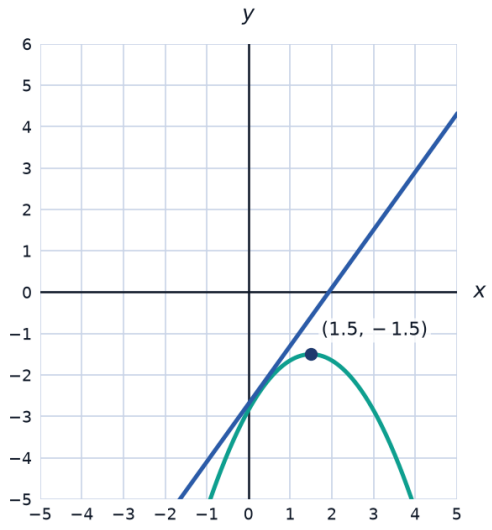
## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

In a manufacturing batch, 5% of components are defective. A sample of 12 components is inspected. Find the probability that at least 2 components are defective.

**16** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Functions: Composite & Inverse

A fair coin is tossed 10 times. Find the probability of getting (a) exactly 6 heads, (b) fewer than 3 heads.




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**17** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Circular Measure

[Binomial Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

**18** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Kinematics

Practice 4.1: A multiple-choice examination consists of 20 independent questions, each having 4 options of which only one is correct. A student guesses randomly on all questions. Let $X$ be the number of correct answers. (a) State the distribution of $X$ including its parameters. (b) Find the probability that the student passes by scoring at least 10 correct answers. (c) Calculate the mean and standard deviation of $X$.

**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 12.01 Forces & Equilibrium

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term $a_1$, and compute the exact value of the partial sum $S_4$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An arithmetic progression has first term $a = 7$ and common difference $d = 4$. Find the 20th term $u_{20}$.

**a.** 83

**b.** 87

**c.** 79

**d.** 80

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum of the first 25 terms of the AP: $3, 8, 13, 18, \ldots$

**a.** 1575

**b.** 1625

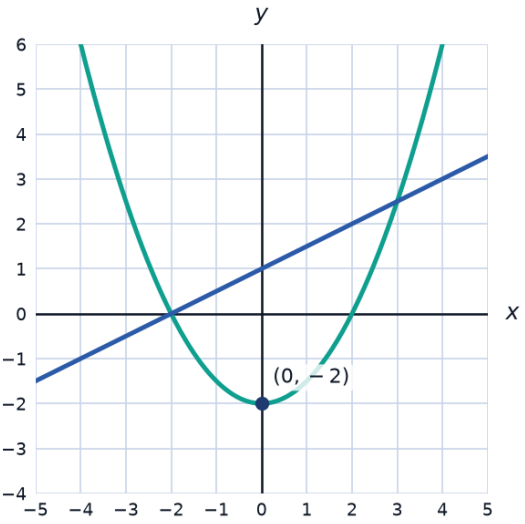
**c.** 1500

**d.** 1550

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**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In an AP, the 5th term is 18 and the 12th term is 39. Find the first term $a$ and common difference $d$.




**a.** $a = 6, d = 3$

**b.** $a = 3, d = 3$

**c.** $a = 9, d = 3$

**d.** $a = 6, d = 4$

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A geometric progression has first term $a = 5$ and common ratio $r = 2$. Find the 8th term $u_8$.

**a.** 640

**b.** 1280

**c.** 320

**d.** 512

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum of the first 6 terms of the GP: $3, -6, 12, -24, \ldots$

**a.** $-63$

**b.** $63$

**c.** $-127$

**d.** $-65$

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The 2nd term of a GP is 12 and the 5th term is $324$. Find the common ratio $r$.

**a.** 3

**b.** 2

**c.** 4

**d.** $\sqrt{3}$

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**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the condition for the sum to infinity of a geometric progression to exist.

**a.** $|r| < 1$

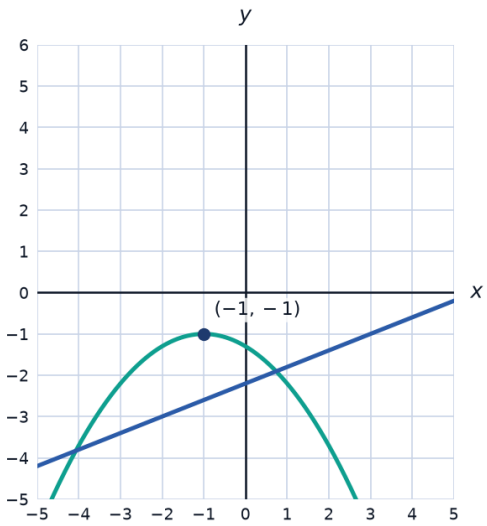
**b.** $r > 1$

**c.** $r \leq 1$

**d.** $|r| > 1$

**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum to infinity of the GP: $16, 12, 9, 6.75, \ldots$




**a.** 64

**b.** 48

**c.** 32

**d.** 72

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Express the recurring decimal $0.\overline{27} = 0.272727\ldots$ as an exact fraction in lowest terms using an infinite geometric series.

**a.** $\frac{3}{11}$

**b.** $\frac{27}{100}$

**c.** $\frac{9}{33}$

**d.** $\frac{7}{25}$

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**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the coefficient of $x^3$ in the expansion of $(2 + 3x)^5$.

**a.** 1080

**b.** 720

**c.** 540

**d.** 1440

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the term independent of $x$ in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

**a.** 240

**b.** 60

**c.** 160

**d.** 120

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the expansion of $(1 + ax)^6$, the coefficient of $x^2$ is 60. Given $a > 0$, find $a$.

**a.** 2

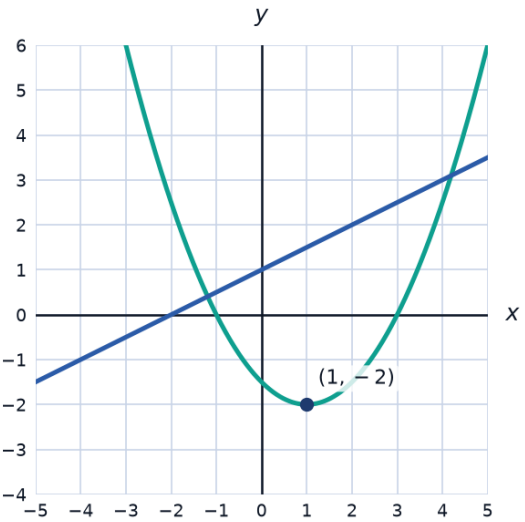
**b.** 4

**c.** 3

**d.** 5

**33** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A geometric series has first term $a = -10$ and common ratio $r = 1/2$. Find $S_6$, the sum of the first 6 terms.




**a.** -315/16

**b.** -155/8

**c.** -475/16

**d.** -635/32

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**34** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the sum of the first $40$ terms of an arithmetic sequence with first term $-11$ and common difference $-2$.

**a.** -2000

**b.** -2004

**c.** -1994

**d.** -2003

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An arithmetic sequence has first term $a = 7$ and common difference $d = 3$. Find the $7$th term.

**a.** 21

**b.** 25

**c.** 24

**d.** 26

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The $4$th term of an arithmetic sequence is $19$, and the common difference is $3$. Find the first term.

**a.** 8

**b.** 9

**c.** 15

**d.** 10

**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An arithmetic sequence has first term $6$ and common difference $5$. Find $n$ such that $S_n = 1403$.

**a.** 19

**b.** 25

**c.** 27

**d.** 23

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A geometric series has first term $a = 11$ and common ratio $r = 1/2$. Find $S_\infty$.

**a.** 11/2

**b.** 22

**c.** 22/3

**d.** 33

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**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Kinematics

The $12$th term of an arithmetic sequence is $58$, and the first term is $14$. Find the common difference.

**a.** 4

**b.** 1

**c.** 6

**d.** -1

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 12.01 Forces & Equilibrium

A geometric sequence has first term $a = 8$ and common ratio $r = 1/3$. Find the $4$th term.

**a.** -8/27

**b.** 8/27

**c.** 8/81

**d.** 8/9

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 13.01 Newton's Laws

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is $p$. The number of defective light bulbs in a random sample of $n$ bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is $2$. \\ (i) Calculate the value of $p$. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly $k$ defective bulbs, where $k$ is the smallest integer such that $P(X \leq k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of $n'$ such that the probability of observing at least one defective bulb is greater than $0.999$ .

**a.** 0.04

**b.** 0.08

**c.** 0.12

**d.** 0.16

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 14.01 Energy, Work & Power

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

**a.** $B(4, 1/6)$

**b.** $B(6, 1/4)$

**c.** $B(1, 4/6)$

**d.** $B(4, 4/6)$

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**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 15.01 Representation of Data

A die is rolled 15 times; $X$ is the number of sixes. Is $X$ binomially distributed?

**a.** No — trials are not independent / probability is not constant

**b.** Yes — but only if $n > 30$

**c.** Yes — fixed, independent trials with constant success probability

**d.** No — the number of trials is not fixed

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 16.01 Permutations & Combinations

A fair coin is tossed 20 times; $X$ is the number of heads. Is $X$ binomially distributed?

**a.** No — trials are not independent / probability is not constant

**b.** Yes — but only if $n > 30$

**c.** Yes — fixed, independent trials with constant success probability

**d.** No — the number of trials is not fixed

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 17.01 Discrete Random Variables

Cards are dealt from a single shuffled deck without replacement; $X$ is the number of aces dealt. Is $X$ binomially distributed?

**a.** No — trials are not independent / probability is not constant

**b.** Yes — but only if $n > 30$

**c.** Yes — fixed, independent trials with constant success probability

**d.** No — the number of trials is not fixed

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